Is the Web App Your AI Built Really Secure?

h5i is a Rust web framework for building apps whose critical rules you can verify, with tools to test the running system.

FRAMEWORK / RUST + AXUM

PROOFS / LEAN 4

TOOLS / DEVELOPMENT INFRASTRUCTURE

Software security has two sides

AI can ship an app quickly. Security still needs both kinds of evidence.

RED-TEAMING

Find Bugs ∃

Find a counterexample in the running application.

FORMAL VERIFICATION

Prove Properties ∀

Prove a stated rule for the modeled logic under explicit assumptions.

h5i: The Verifiable Web Framework

CORE PRODUCT

Verifiable Web Framework

Build on Axum with composable Rust kernels. Extract application logic to Lean 4 to prove critical rules, with optional GPU acceleration for suitable workloads.

RUST / AXUM | LEAN 4 | OPTIONAL GPU

INTEGRATED TOOLING

Development Infrastructure

A unified CLI for agent-led red-teaming, browser and HTTP automation, security regression tests, and sandboxed workflows.

Works with existing web applications, too.

Write Once. Verify with Lean 4.

Axum and storage stay outside the pure Rust kernel, so application decisions can be extracted and proved. GPU execution is an optional optimization.

Rust / parallel/kernel/src/lib.rs · pure kernel

```rust
pub fn push_rule(event: &[u8], pattern: &[u8],
    display: &[u8], members: u64,
    minimum: u64, maximum: u64, power: u64,
    required: u64, enabled: bool,
    require_display: bool) -> u32 {
    if !enabled { return 0; }
    if power < required { return 0; }
    if members < minimum
        || members > maximum { return 0; }
    if !glob(event, pattern) { return 0; }
    // display-name check omitted
    1
}
```

No I/O and no allocation: Axum and storage adapters stay outside.

Charon
Aeneas

→

Lean 4 / Proofs.lean · verify

```
theorem insufficient_power_denies ...
    (h : power < required) :
    push_rule event pattern display members
      minimum maximum power required true
      requireDisplay = ok 0#u32 := by
  simp [push_rule, h]
```

Rust-CUDA
PTX

→

CUDA / device/src/lib.rs · optional

```rust
#[kernel]
pub unsafe fn push_batch(...) {
    let i = thread::index_1d() as usize;
    ... shared::push_rule(&events[row(i)], ...)
}
```

15.6× 1M push-rule rows vs one CPU thread · RTX 3090

Same kernel source on CPU, in Lean, and on GPU / GPU backend is experimental; Lean does not prove CPU/GPU equivalence

Prove Critical Application Properties

Rust / kernel/src/lib.rs

```rust
fn delete(user: u64, s: &Snapshot,
          id: u64) -> Outcome {
    match find_post(&s.posts, id) {
        None => Err(Error::NotFound),
        Some(p) => {
            if p.author == user
                || is_moderator(&s.moderators, user) {
                Ok((one(Write::DelPost(id)),
                    Reply::Done))
            } else {
                Err(Error::Forbidden)
            }
        }
    }
}
```

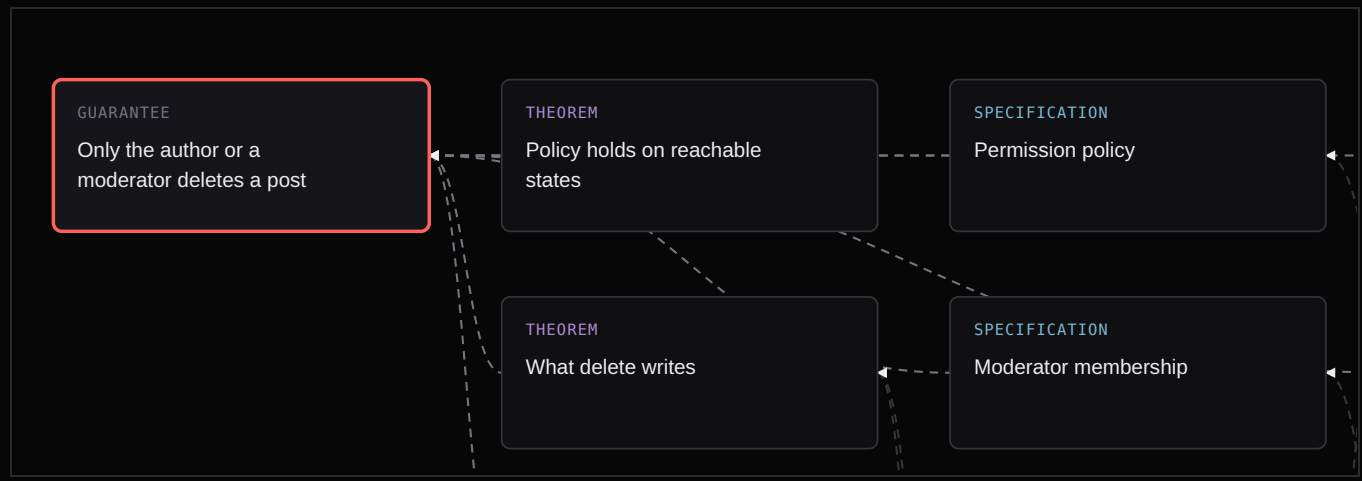
Charon
Aeneas

→

Lean 4 / Theorems.lean · excerpt

```
theorem authorized (a : Principal) (s : Snapshot)
    (c : Command) ws r
    (hinv : Inv (Snapshot.toSt s))
    (h : transition a s c = .ok (.Ok (ws, r))) :
    ∀ w ∈ ws.val, allowed (Snapshot.toSt s) a.user.val w := by -- proof body omitted
```

h5i ui / theorem-spec dependency graph · excerpt



Proofs apply to stated properties under documented assumptions

Development Infrastructure

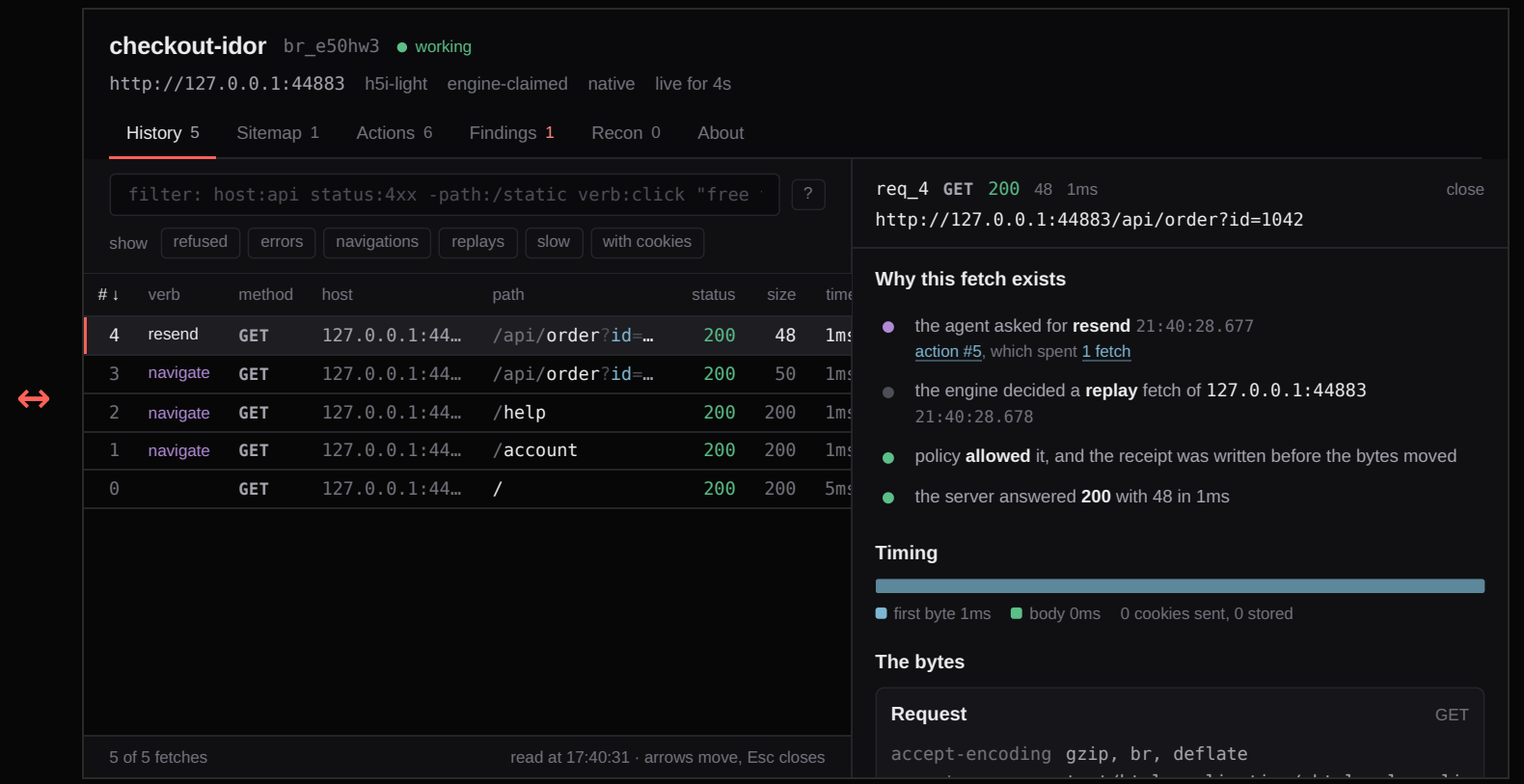
Red-teaming, CI/CD testing, and sandboxed workflows for h5i-app or any existing web application.

AI agent / shell

```
$ h5i browser snapshot
  @e7 Order 1041

$ h5i websec replay req_3 \
    --set query.id=1042
  → req_4 · HTTP 200

$ h5i websec diff res_3 res_4
  owner: Alice → Bob
```



Illustrative shell + actual h5i ui / local demo; confirmed findings can become CI tests

07 / EARLY TRACTION

Open-source traction

2.1k+

RELEASE DOWNLOADS

790+

GITHUB STARS

Maintainer-reported · October 21, 2026 Repository ↗ Releases ↗

Build and Verify Your Web Apps.

We're looking for developers and teams building security-critical web applications to help shape the next generation of verifiable web development.

Start a conversation ↗

Explore h5i ↗